

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \langle \lambda_s(b)\xi, \eta \rangle &= \sum_{t \in G} \lambda_s(b)\xi(t)^* \eta(t) = \sum_{t \in G} (b\xi(s^{-1}t))^* \eta(t) = \sum_{t \in G} \xi(s^{-1}t)^* b^* \eta(t) \\ &\stackrel{r=s^{-1}t}{=} \sum_{r \in G} \xi(r)^* b^* \eta(sr) = \sum_{r \in G} \xi(r)^* \lambda_{s^{-1}}(b^*) \eta(r) = \langle \xi, \lambda_{s^{-1}}(b^*) \eta \rangle. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\sum_{i=0}^{m-1} \{ \sum_{i'=0}^{l_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{i'}(g(\sigma_B^i(h(x)))))) - \sum_{j'=0}^{k_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{j'}(g(\sigma_B^i(h(x)))))) \} \\ &= \sum_{i'=0}^{l_2^m(h(x))-1} f(\sigma_C^{i'}(gh(x))) - \sum_{j'=0}^{k_2^m(h(x))-1} f(\sigma_C^{j'}(g(\sigma_B^m(h(x))))). \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$P\psi = P \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \psi_4 \end{pmatrix}; \quad P_\mu \psi = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \psi_\mu \end{pmatrix} \quad (\mu = 0, \dots, 3).$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \lim_{r \rightarrow 0^+} \frac{A(r)^2}{A'(r)} \cdot [\mathcal{W}(r)^{p-1-\alpha}]' &= \lim_{r \rightarrow 0^+} A(r) \cdot [Q_1(\mathcal{V}(r), v_0(r)) - \mathcal{W}(r)^{p-1-\alpha}] \\ &= \lim_{r \rightarrow 0^+} [r^\delta u'_0(r)^{p-1-\alpha} Q_1(\mathcal{V}(r), v_0(r)) - r^\delta u'(r)^{p-1-\alpha}] \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
\bar{P}_{j,j+i} = & \sum_{m=i+1}^{M-j} \binom{M-j}{m} \mathbb{P}_{\text{TX}}^m (1 - \mathbb{P}_{\text{TX}})^{M-j-m} \\
& \times m \cdots (m-i+1) \mathbb{P}_K^m \gamma_i \gamma_{m,i} \\
& + \binom{M-j}{i} \mathbb{P}_{\text{TX}}^i (1 - \mathbb{P}_{\text{TX}})^{M-j-i} i! \mathbb{P}_K^i \gamma_i
\end{aligned}$$